Dwarkadas J. Sanghvi College of Engineering

B.Tech in Artificial Intelligence (AI) and Data Science, Honors in Business Intelligence

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Strategic Business Intelligence and Decision Engineering (DJS23SH2301L)

Mini Project

An End-to-End Strategic BI Solution for an Online Retailer

KPI scorecard, exception alerts, customer analytics, what-if simulation and vendor selection

Submitted by: [Your name]

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Code: github.com/vihanlalan/bi-miniproject

Abstract

This mini project builds a complete strategic BI solution for ShopSphere, a simulated online retailer with 22,739 order lines, 40 products and 6 vendors over 24 months. Raw files are cleaned and loaded into a star schema in SQLite. On top of that I built a balanced scorecard, an exception-alert engine, RFM customer segmentation, a channel-level CLV against CAC comparison, a churn decision tree, a what-if and Monte Carlo model for profit, and a TOPSIS ranking of vendors. The results are presented in a six-page Power BI report with DAX measures and a live what-if page, and also as an HTML dashboard and a Streamlit workbench.

For FY2025 the retailer earned Rs 5.53 crore (92% of budget) at a gross margin of 25.1%, against a 28% target. The system flagged two problems that were planted in the data: Social-media acquisition cost rising from about Rs 1,886 to Rs 5,063 per customer, and one vendor falling to 20% on-time delivery. The what-if model shows that a 5% rise in supplier cost would remove 20% of contribution, far more than any marketing change could recover.

All data is synthetic. The findings show that the method works. They are not claims about a real company.

1. Introduction

1.1 Business context and problem

ShopSphere sells electronics, home and kitchen goods, fashion, beauty products and sports gear online. Its management team gets revenue reports every month, but nobody connects them to the strategy. Three questions come up again and again. Are we meeting our targets, and where are we not? Which customers and channels deserve the next rupee? And if we change price, discounts or marketing, what happens to profit and how sure can we be? A set of operational reports cannot answer these. Strategic BI has to.

1.2 Objectives

- Build a clean, tested data warehouse that every KPI reads from, so two reports never give two answers.
- Translate strategy into KPIs with targets and a balanced scorecard (Unit 3).
- Detect problems early with rule-based and statistical alerts instead of waiting for a monthly review (Unit 3).
- Cover the finance, marketing and operations functions with analytics that lead to a decision (Unit 4).
- Support decisions under uncertainty with what-if, goal seek, sensitivity, Monte Carlo and a multi-criteria vendor model (Unit 2).
- State the governance and ethical limits of the solution (Unit 6).

1.3 Scope

I picked finance, marketing and operations from Unit 4. HR analytics is left out because a retail dataset of this kind would need invented employee data, which adds nothing. I used a Python-only pipeline for the data preparation and the heavier models (ETL, segmentation, churn tree, TOPSIS, Monte Carlo), because those steps are then reproducible and tested. The presentation layer is Power BI Desktop: the star schema is loaded from one Excel workbook, KPIs and the what-if projection are DAX measures, and the report has six pages (Section 3.4).

2. Mapping to the syllabus

The table shows where each syllabus unit appears in the project.

Unit	Syllabus topic	What was built	File
1	Strategic vs operational decisions, BI maturity, case studies	Business case, scorecard that links operational data to strategic targets	bi/kpi.py
2	Decision trees, what-if, goal seek, sensitivity, risk, MCDM, simulation	Churn decision tree, what-if levers, goal seek, tornado chart, Monte Carlo, TOPSIS and weighted sum	bi/scenario.py, bi/mcdm.py, bi/segmentation.py
3	EPM, KPIs, balanced scorecard, dashboard design, alerts, exception reporting	Nine-KPI balanced scorecard with RAG status, alert engine, executive overview page in Power BI (HTML copy also kept)	powerbi/measures.dax, bi/alerts.py
4	Finance, marketing and operations analytics	Budget variance and margin analysis, RFM, CLV and CAC by channel, vendor performance	bi/kpi.py, bi/segmentation.py
5	Self-service BI, automated insights, real-time analytics	Power BI what-if page with slicers as parameters, and a Streamlit copy. Publishing to Power BI Service and Q&A are described in the build guide.	powerbi/, app.py
6	Data quality, governance, ethics, ROI, mini-capstone planning	Cleaning log, post-load checks, 12 automated tests, governance notes in Section 9	bi/etl.py, tests/

3. Data and architecture

3.1 The synthetic dataset

I generated the data with a fixed random seed so that anyone running the code gets the same numbers. Customers sign up through five channels (Search, Social, Email, Affiliate and Organic) and place orders month by month with seasonality, a promotion in June, November and December, and a gradual loss of customers over time. Vendor lead times, defect rates and cost levels differ, which is what makes vendor selection a real decision. The budget is the actual run rate with a planning error of -7% to +10% per month, which mimics a finance team that plans without knowing about later problems.

Two problems were planted on purpose so that the BI layer has something to find. From September 2025, 55% of Social-media sign-ups disappear while the spend stays the same. From October 2025, deliveries from one vendor, Evergreen Imports, take about three days longer. A small amount of dirty data was added as well: duplicate rows, missing delivery days and negative quantities.

3.2 Pipeline and star schema

The pipeline has four steps: generate raw CSV files, clean and load them into SQLite, run the analytics, and write the dashboard, figures and results file. Running `python -m bi.pipeline` does all four in about 45 seconds. The warehouse is a star schema with one sales fact table (one row per order line) and dimensions for date, customer, product and vendor. Marketing spend and the budget sit in two smaller fact tables at month level. Measures such as net revenue, cost of goods, return cost and gross profit are calculated once during loading, so no report can define them differently.

3.3 Data quality

The cleaning rules are simple and each one has a reason. Duplicate line IDs are dropped because the same sale would otherwise be counted twice. Rows with a negative quantity are dropped because returns are already tracked in a separate flag. Missing delivery days are filled with the median for the same product, since delivery time depends mostly on the vendor of that product. After loading, five checks run (null keys, duplicate IDs, orphan customers, orders dated before sign-up, negative revenue) and all must return zero.

Step	Rows affected
Raw order lines	22,796
Duplicate lines removed	45
Negative quantities removed	12
Missing delivery days filled with product median	70
Clean order lines loaded	22,739
Post-load checks failing	0

3.4 Power BI implementation

The warehouse tables and the model results are exported to one Excel workbook with 17 sheets, which Power BI loads in a single step. The model has nine relationships, all many-to-one with single-direction filtering: a month table filters the date table, the date, customer and product tables filter sales, vendors filter products, and a channel table filters both customers and marketing spend. A separate month table is needed because marketing spend and the budget are monthly, while sales are daily, and a shared month key lets one slicer drive all three.

About 70 DAX measures cover revenue, margin, return rate, on-time delivery, budget variance, CAC and contribution. The headline FY2025 measures use REMOVEFILTERS on the date tables so that they do not change when someone clicks a month, which is how an executive card should behave. The what-if page uses seven calculated tables built with GENERATESERIES as slicers. A measure reads each slicer with SELECTEDVALUE and falls back to a baseline value when nothing is selected, so the page opens showing the actual FY2025 position. The projection measure uses the same formula as the Python model, and the build guide lists the Python results to check the DAX against.

Page	Content	Syllabus link
1. Executive overview	KPI cards, revenue against budget, margin trend, exception table	Unit 3
2. Balanced scorecard	Nine KPIs by perspective with red, amber and green status	Unit 3
3. Marketing and customers	CAC by channel, RFM segments, CLV against CAC, churn risk	Unit 4
4. Operations and vendors	On-time trend by vendor, TOPSIS ranking, margin by category	Units 2 and 4
5. What-if analysis	Seven slicers, projected against baseline contribution, preset scenarios	Unit 2
6. Risk and sensitivity	Tornado chart, Monte Carlo histogram and percentiles	Unit 2

4. Strategic KPIs and the balanced scorecard

I chose nine KPIs across the four perspectives of Kaplan and Norton's balanced scorecard. Each has a target, and a status of Green (target met), Amber (within 90% of target) or Red. For KPIs where lower is better, such as return rate, attainment is the target divided by the actual value. For FY2025, 5 KPIs are

Green, 3 are Amber and 1 is Red.

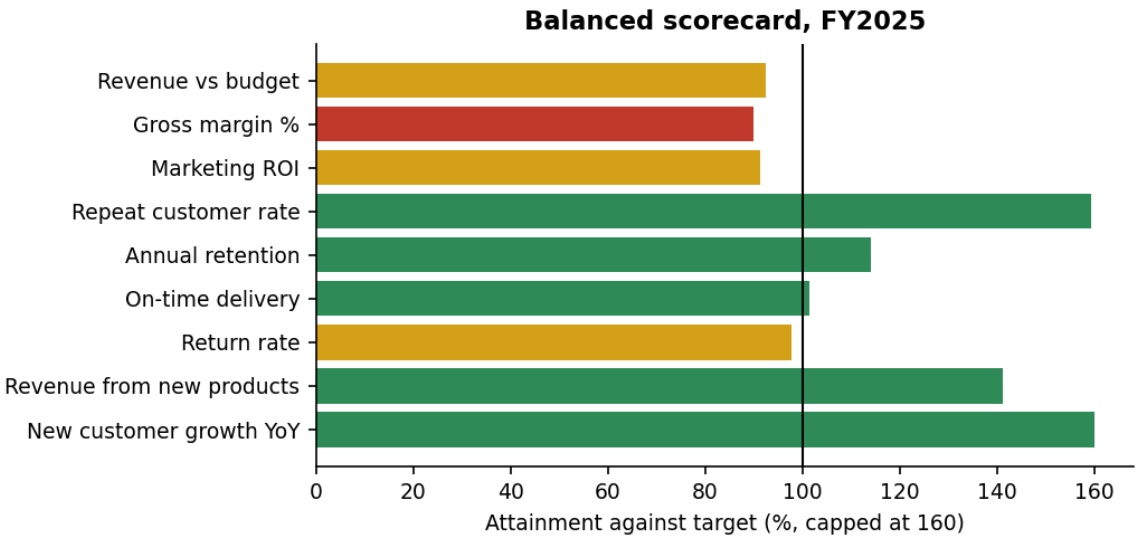


Figure 1. Attainment against target for each KPI.

Perspective	KPI	Actual	Target	Status
Financial	Revenue vs budget	92.4%	100.0%	Amber
Financial	Gross margin %	25.1%	28.0%	Red
Financial	Marketing ROI (first-year gross profit per rupee of spend)	1.83	2.00	Amber
Customer	Repeat customer rate	71.7%	45.0%	Green
Customer	Annual retention	57.0%	50.0%	Green
Internal process	On-time delivery	86.2%	85.0%	Green
Internal process	Return rate (by value)	8.2%	8.0%	Amber
Learning & growth	Revenue from new products	21.2%	15.0%	Green
Learning & growth	New customer growth YoY	31.3%	10.0%	Green

The only Red KPI is gross margin at 25.1%. The cause is visible in the monthly chart. In the three promotion months the margin is 17.9%, against 28.7% in the other nine months. A 10 to 20% discount removes most of the thin margin on electronics, which is 47% of revenue and earns only 14.4% gross margin. So the promotion calendar, and not the cost base, is what holds the margin below target.

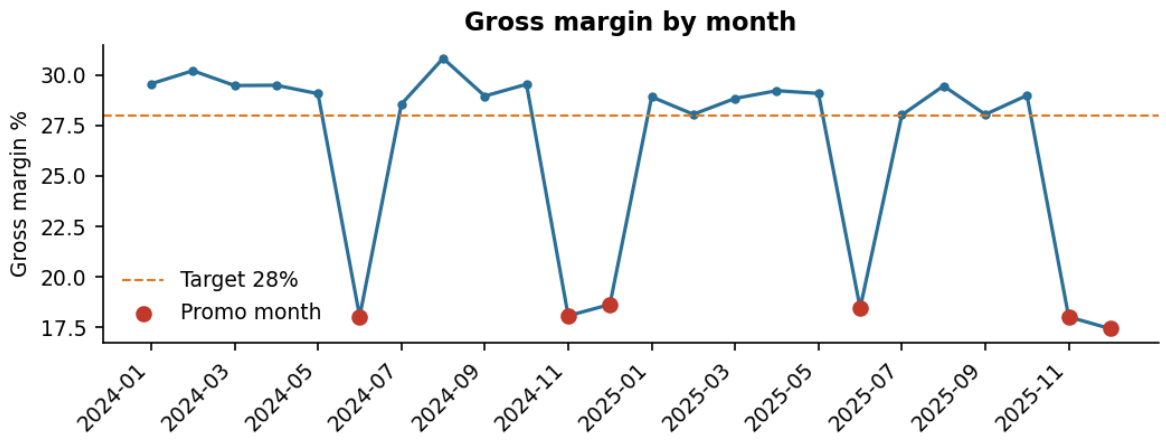


Figure 2. Gross margin by month. Red dots are promotion months.

Revenue reached 92% of budget even though it grew 126% on last year. The first caveat is that the budget here is a simulated plan, so this gap says little about real forecasting. The second is that Amber on marketing ROI (1.83 against a target of 2.0) is partly built in: I count only first-year gross profit from customers acquired in 2025, and a customer who joins in July has had six months to pay back.

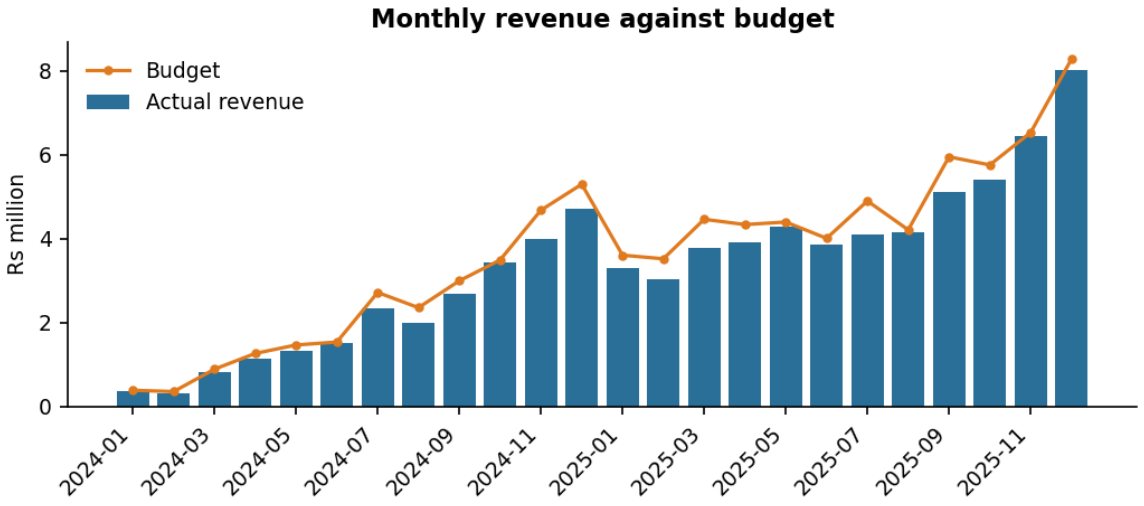


Figure 3. Monthly revenue against budget, Rs million.

5. Performance monitoring and exception alerts

A dashboard that someone must open and read is passive. The alert engine looks at the latest three months and raises an exception when one of these rules fires: revenue under 90% of budget, gross margin more than three points under target, marketing spend more than 10% over budget, a vendor under 80% on-time delivery, or a channel's customer acquisition cost more than two standard deviations above its own previous 12 months. The z-score rule matters because a fixed threshold has no way to know that Rs 2,500 is normal for Search and a disaster for Email.

Month	Severity	Area	Alert
2025-12	High	Vendor	Evergreen Imports: on-time delivery 20%
2025-11	High	Marketing	Social CAC is Rs 4,902, z-score 2.3 vs the prior 12 months

Month	Severity	Area	Alert
2025-10	High	Marketing	Search CAC is Rs 2,624, z-score 2.2 vs the prior 12 months
2025-10	High	Marketing	Social CAC is Rs 5,063, z-score 3.8 vs the prior 12 months
2025-12	Medium	Margin	Gross margin 17.4% is under target
2025-12	Medium	Vendor	Fulcrum Goods: on-time delivery 77%
2025-11	Medium	Margin	Gross margin 18.0% is under target

Both planted problems were caught without anyone looking for them. Social acquisition cost went from Rs 1,886 in August to Rs 4,670 in September and Rs 5,063 in October, with a z-score of 3.8 in October. Evergreen Imports dropped from 96% on-time in September to 19% in October. That vendor supplies 19% of revenue, so the delay is not a small issue. The Search alert in October is a smaller side effect: Search CAC in October was Rs 2,624, a little above its usual range.

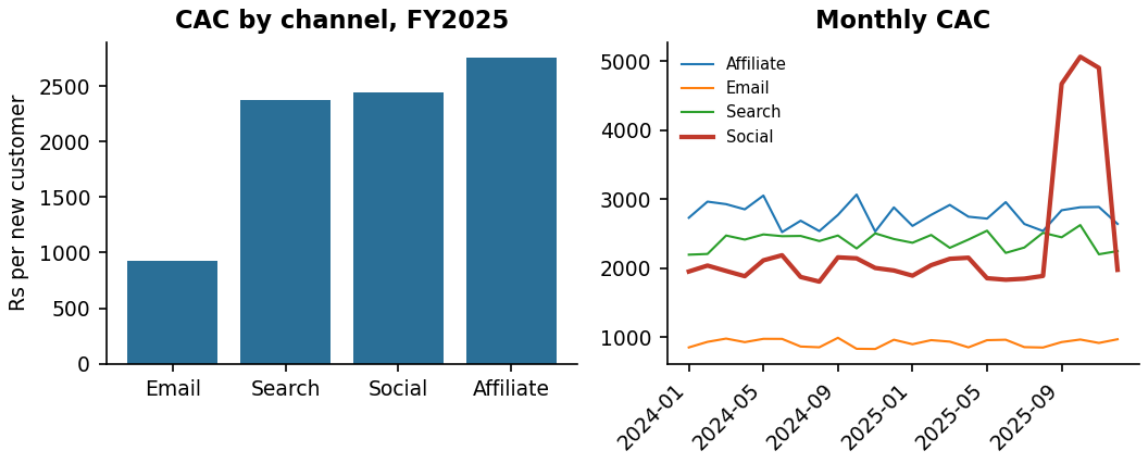


Figure 4. CAC by channel for FY2025 (left) and monthly CAC (right). The Social spike is the planted problem.

6. Functional analytics

6.1 Finance: where the margin comes from

Category margins range from 14.4% for Electronics to 46.7% for Beauty. Electronics brings the most revenue (Rs 259.4 lakh) but only Rs 37.4 lakh of gross profit, while Beauty brings Rs 31.7 lakh on about a quarter of Electronics' revenue. Fashion has the highest return rate at 16.5%, more than three times the rate of Beauty. Any promotion budget should be set per category, because the same discount has very different effects on each.

6.2 Marketing: segments, channels and churn

I scored each customer from 1 to 5 on recency, frequency and monetary value (RFM) and grouped them with simple rules. The result is uneven in the way retail usually is. Champions are 19% of customers and bring 38% of revenue. The top 20% of customers by spend account for 55% of revenue. The group that needs attention is At risk: 540 customers who used to buy often (5.8 orders on average) but have not bought for about 322 days. They represent 28% of lifetime revenue, so a win-back campaign aimed at them is worth more than the same effort on new customers.

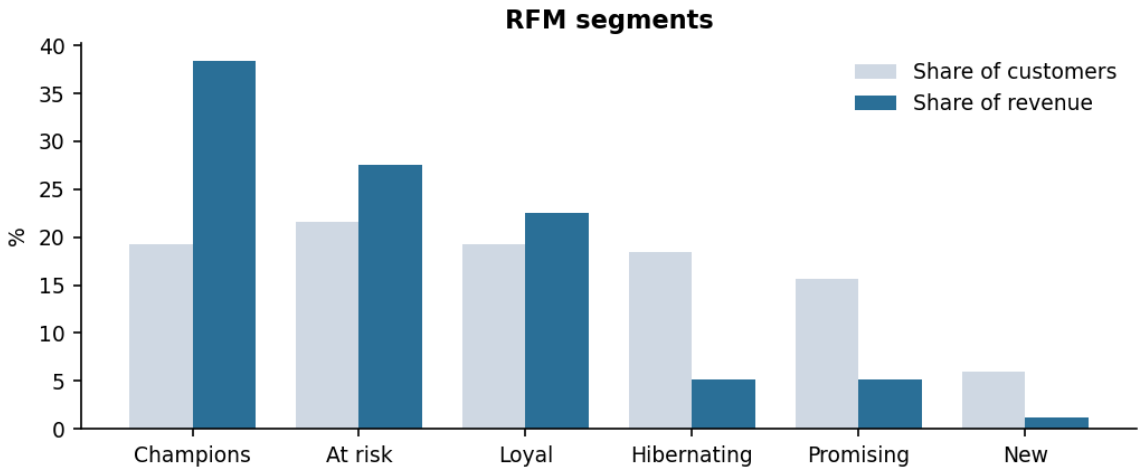


Figure 5. Share of customers and share of revenue by RFM segment.

To compare channels I used a simple customer lifetime value formula, CLV = m x r / (1 + i - r), where m is the yearly gross profit per active customer, r is the share of last year's customers who bought again this year, and i is a 10% discount rate. Dividing by the channel's CAC gives a payback ratio. A common rule of thumb is that this ratio should be at least 3.

Channel	Retention	CLV (Rs)	CAC (Rs)	CLV / CAC
Email	60%	7,756	922	8.4
Social	59%	8,073	2,444	3.3
Affiliate	59%	8,182	2,758	3.0
Search	54%	6,469	2,374	2.7
Organic	55%	6,425	n/a	n/a

Email returns 8.4 times its cost, far above the others. Search is the weakest paid channel at 2.7 and Affiliate is just under the threshold. Social looks fine on this annual view (3.3), but only because the three bad months are averaged with nine good ones. This is why the monthly alert is needed alongside the annual metric.

For churn I trained a decision tree of depth 3 on customers as of 31 March 2025 and checked who bought nothing in the next 180 days. I chose a tree over a more accurate model because a marketing manager can read its rules. On a held-out 30% of customers it reaches 71% accuracy against 53% for always guessing the majority class, with an AUC of 0.79. Recency does almost all the work (92% of importance): customers who have not bought for roughly 90 days are very likely to stay quiet. That is useful but not surprising, and a single split on one date is not enough to claim more, so I treat the model as a ranking aid and not as a forecast.

6.3 Operations: choosing vendors

Vendors differ on cost, delivery and quality, and none is best on all three. I ranked them on four criteria, cost index, on-time rate, return rate and average delivery days, with weights of 30%, 30%, 25% and 15%. TOPSIS scores each vendor by its distance from an ideal vendor and a worst vendor. I also ran a plain weighted sum as a cross-check, because two methods that agree are more convincing than one.

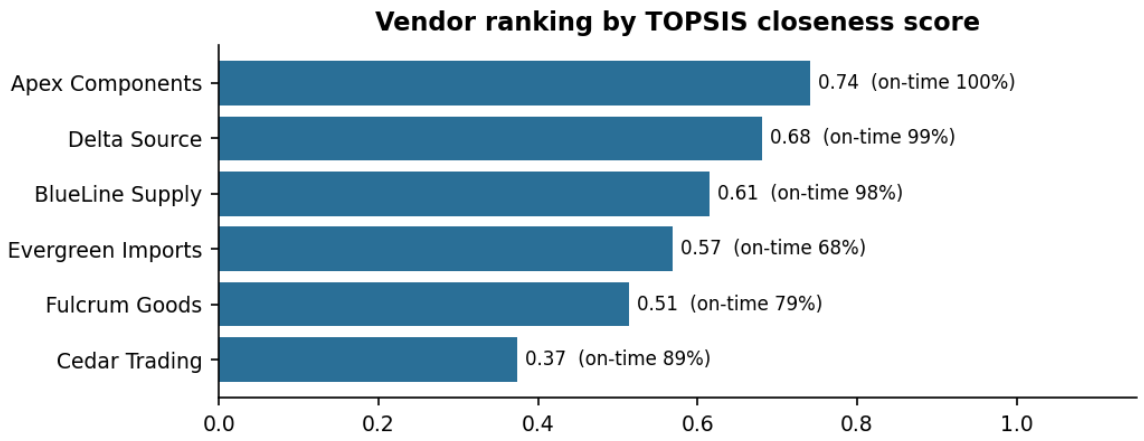


Figure 6. TOPSIS score for each vendor, with FY2025 on-time rate.

Apex Components is first under both methods. Evergreen Imports is fourth under TOPSIS and sixth under the weighted sum, because its low price of 0.88 is cancelled out by 68% on-time delivery. Cedar Trading is last under TOPSIS because of a 12% return rate. Since the weights are a judgement call, I drew 5,000 random weight sets and counted how often each vendor ranks first. Apex wins 56% of the time and Delta Source 20%, so the top choice is fairly stable but not certain.

7. Decision engineering: what-if, sensitivity and risk

7.1 The model

The model projects next year's contribution (gross profit minus marketing) from the FY2025 actuals. It has five levers: list price, discount depth, marketing spend, return rate and supplier cost. Demand responds to the effective price with a constant elasticity of -1.3, and to marketing with an exponent of 0.6 applied to the share of revenue that comes from newly acquired customers, so that extra spend has diminishing returns. Both numbers are assumptions and not estimates from the data, which is why they are inputs in the workbench and are varied in the Monte Carlo run. With no lever moved, the model reproduces the actual FY2025 contribution to within 1%, and a test checks this.

7.2 Scenarios

Scenario	Revenue (Rs crore)	Contribution (Rs crore)	Revenue change	Contribution change
Baseline (no change)	5.53	1.03	+0.0%	+0.0%
Price +3%	5.48	1.13	-0.9%	+10.2%
Price -3%	5.58	0.91	+0.9%	-11.1%
Marketing +20%	5.90	1.05	+6.7%	+2.0%
Marketing -20%	5.13	1.00	-7.3%	-2.8%
Discount +3 pts	5.59	0.91	+1.0%	-11.9%
Supplier costs +5%	5.53	0.83	+0.0%	-19.8%
Cut returns by 2 pts	5.65	1.08	+2.2%	+4.9%
Combined: price +2%, marketing +10%, returns -2 pts	5.81	1.17	+5.0%	+13.4%

A 3% price rise lowers revenue by 0.9% but raises contribution by 10.2%, because the 3.8% of units lost no longer need to be bought from vendors, while every remaining unit earns 3% more. A 3-point deeper discount has the opposite effect and costs 11.9% of contribution. Marketing is the weakest lever: 20% more spend adds 6.7% revenue but only 2.0% contribution, since marketing is a small share of cost. The largest threat is outside the company's control: a 5% rise in supplier cost reduces contribution by 19.8%.

7.3 Sensitivity and goal seek

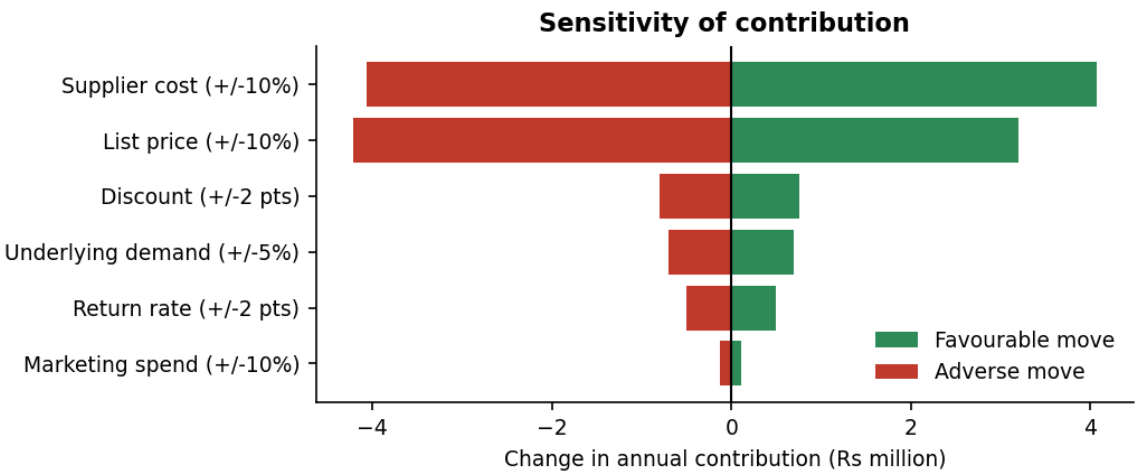


Figure 7. Change in annual contribution when each driver moves by the stated amount.

The tornado chart ranks the drivers. Supplier cost and list price are far ahead of everything else, each swinging contribution by about Rs 81.3 lakh over the tested range, while a 10% change in marketing spend moves it by only Rs 2.5 lakh. Management attention should follow that order. Two goal-seek results follow from the model. To lift contribution by 10%, list prices would need to rise by 2.9% if nothing else changes. And a 3% price rise stays profitable unless customers are much more price-sensitive than assumed: the break-even elasticity is -3.76, against the assumed -1.3.

7.4 Monte Carlo risk

The point estimates above hide uncertainty. I ran 10,000 simulations of the combined plan (price +2%, marketing +10%, returns down 2 points) with elasticity drawn from a normal distribution around -1.3, the marketing exponent between 0.4 and 0.8, a demand shock of 6% standard deviation, and supplier cost inflation averaging 2%. The mean contribution is Rs 1.09 crore, with a 5th to 95th percentile range of Rs 0.84 crore to Rs 1.35 crore. The do-nothing baseline is Rs 1.03 crore, so the plan beats it in about 63% of runs. That number is lower than the deterministic +13.4% suggests because the simulation includes expected cost inflation that the baseline does not. A decision maker should read the range and not the single figure.

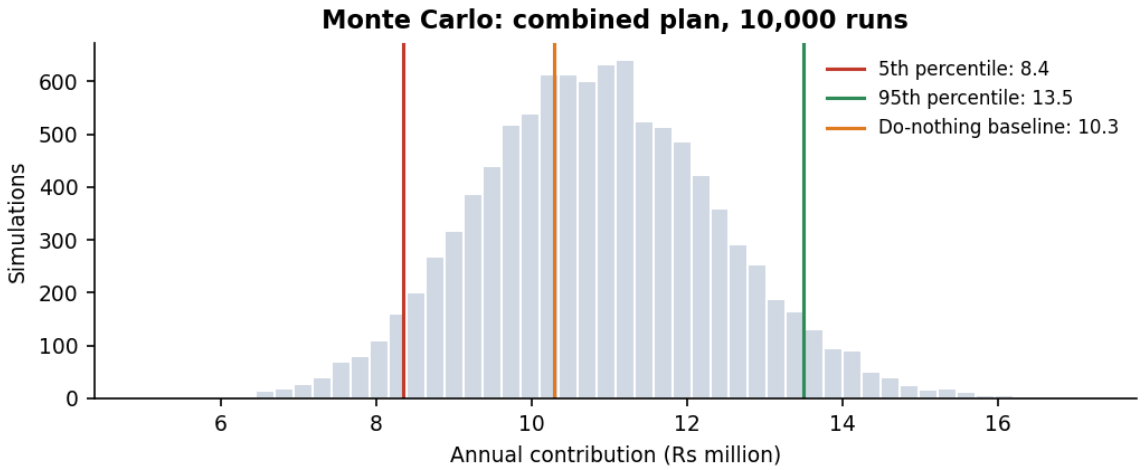


Figure 8. Distribution of annual contribution under the combined plan.

8. Recommendations

- **Vendors.** Open a service-level discussion with Evergreen Imports now, since its on-time rate fell from 96% to 20% and it supplies 19% of revenue. Start moving some volume to Apex Components and Delta Source, which rank first and second.
- **Acquisition.** Pause or cap Social spend whenever its CAC z-score passes 2, and shift part of the budget to Email, which returns 8.4 times its cost. Email has limited reach, so test how far it scales.
- **Promotions.** Set discount depth per category. Electronics earns 14.4% margin and cannot carry a 15% discount. This is the main route to the 28% margin target.
- **Retention.** Run a win-back campaign for the 540 At-risk customers, who carry 28% of lifetime revenue, before spending more on acquiring new ones.
- **Pricing.** Test a 2 to 3% price rise on a few categories first. The model says it pays unless elasticity is worse than -3.8, but that elasticity has never been measured in this data.

9. Governance, ethics and limitations

9.1 Governance and ethics

Data quality is handled in the load step, with a logged count for every cleaning rule and five post-load checks, so a report can be traced back to a rule instead of someone's judgement. The segmentation uses purchase behaviour only and no personal attributes, and customer IDs are synthetic. In a real deployment I would add role-based access so that marketing sees segments but not individual order history, and I would not use the churn score to decide who gets worse service or worse prices. The decision tree was chosen partly for this reason: its rules can be explained to a customer or an auditor. Vendor weights are visible and editable, so the ranking can be challenged.

9.2 Limitations

- The data is simulated. Planted problems are easier to find than real ones, so the alert results should not be read as a measure of how well the engine works on real data.

- Elasticity and marketing response are assumed, not estimated. The Monte Carlo run shows the effect of being wrong, but cannot remove it.
- The model covers contribution only. There are no fixed overheads, no stock holding costs and no cannibalisation between products.
- The churn model uses one cutoff date and one train-test split. Cross-validation over several cutoffs is the obvious next step.
- Publishing to Power BI Service, Q&A (natural-language query) and HR analytics were not part of the build. The scorecard, alerts, segments and Monte Carlo results are loaded into Power BI as tables from the Python models, not recomputed in DAX.

10. Conclusion

The project takes a retailer's raw files to a set of decisions. The warehouse gives one version of the truth, the scorecard shows that margin is the one KPI in Red, the alert engine finds two problems that a monthly report would have caught late, and the decision models show which levers matter. The most useful result is how unevenly the levers compare: supplier cost and price move profit by tens of lakhs, while marketing moves it by a few lakhs. If I extended the work, I would replace the assumed elasticity with one estimated from price changes in real data, and publish the report through Power BI Service to cover the cloud part of Unit 5.

11. How to run the project

Install the packages with `pip install -r requirements.txt`. Then `python -m bi.pipeline` rebuilds everything and writes `outputs/dashboard.html`. `python -m bi.export_powerbi` writes the Excel workbook for Power BI, and `powerbi/BUILD_GUIDE.md` lists the relationships, measures and visuals for each page. `streamlit run app.py` opens the Streamlit copy of the what-if workbench, `pytest` runs the 12 tests, and `python report/build_report.py` regenerates this report.

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